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TON618 CAPITAL · RESEARCH

The Tripwires: A Six-Week Re-Score

Quantum vs. Crypto, Update One — the scoreboard moved, the stocks moved the other way

TON618 Capital Research · Follow-up to “How Close Is Quantum Computing to Being Useful?” (v1.0, June 28, 2026) · DRAFT — unformatted; adversarial verifier pass applied Aug 6, 2026

> **The score, up front.** In June we closed our first quantum assessment with three pre-registered tripwires — the specific observations that would change our mind about the timeline. Six weeks later we came back to score them, and the period splits three ways. **The lab sped up:** one tripwire is warm — the logical-qubit frontier moved 48 → 70 (claimed), inside the heaviest six-week run of peer-reviewed results the field has ever produced. **The tape gave up:** the four pure-play quantum stocks fell 10–26% through that exact window, 29–43% from their early-June highs. **And the chain hasn’t decided:** Bitcoin’s fix is now funded and specified — and still zero-percent activated. This update re-scores our own scoreboard, line by line, and then re-arms it.

Headline stats (for the stat-tile row): - **1 of 3** — tripwires from v1.0 now partially triggered (logical-qubit counts). - **70** — logical qubits in IBM’s July 30 “verified quantum advantage” claim, vs. 48 at the frontier when v1.0 published. Google’s estimate of what breaks Bitcoin’s curve: ~1,200. - **−17%** — the equal-weight pure-play quantum basket since v1.0’s pricing date (Jun 24, through Aug 6), through the best news cycle the field has ever had. - **$15M / ~$446B** — what the industry has pledged for Bitcoin’s quantum defense, against the market value of the exposed coins.

Executive summary

Answer first: **the timeline is accelerating — modestly, measurably, and on exactly the axes we said to watch — and the equity market is not signaling imminence; if anything it is signaling a hangover.** Those two sentences point in opposite directions, and the distance between them is this note’s subject: the lab and the tape have decoupled, and only one of them is evidence about Q-Day.

Six weeks ago we wrote that quantum computing had produced zero economically useful results, that the error-correction threshold crossing of 2024–25 was real and foundational, and that the honest pacing ran on three clocks — narrow scientific value now, commercial value in the 2030s, code-breaking at roughly one-in-three odds within ten years. We closed with three tripwires that would change our mind. This note goes back and scores them.

The re-score: **one tripwire is partially triggered.** On July 30, IBM and University of Chicago researchers claimed a verified quantum advantage on a computation using 70 logical qubits — a step from “a few dozen” toward the “hundreds” we flagged as the threshold of concern, and the strongest of three quantum-advantage claims IBM coordinated that day. The claim is fresh, the classical-computing community has not yet had its usual turn at bat, and the history of this field is that supremacy claims get caught. But our v1.0 marquee stat — “zero economically useful results” — now carries an asterisk for the first time, and intellectual honesty requires us to print that.

Meanwhile, the market answered the question “are quantum stocks signaling an imminent breakthrough?” with an emphatic no. Since v1.0’s pricing date the four pure-plays fell between 9.5% (QUBT) and 26% (IonQ); from their early-June highs, 29–43%. D-Wave announced a genuine Nature hardware result on August 5 and its stock fell 9% the following session. Even Arqit, the post-quantum-security name we flagged as the “defensive side” of the trade, fell 25%. The pure-play basket is worth roughly $30 billion today against trailing revenue still in the low hundreds of millions — revenue that is, to be fair, growing fast: IonQ alone printed $80 million in Q2, so the sales multiple is compressing from both directions. The de-rating tells you the December–January speculative episode is unwinding; it tells you nothing about Q-Day. Equity prices are a sentiment gauge, not a physics sensor — and the divergence between the news flow and the tape over these six weeks is the cleanest demonstration of that we are ever likely to get.

On Bitcoin specifically, the story of the summer is that the fix moved from “monitorable” to **funded and specified — but not activated.** BIP-360 was merged into the official BIPs repository in February and redesigned along the way into a cleaner construction (Pay-to-Merkle-Root). On July 23, nine institutions — BlackRock, Coinbase, Strategy, Fidelity Digital Assets, Galaxy, Block, Blockstream, Anchorage, and ARK Invest — pledged $15 million over three years to fund the open-source work, explicitly disclaiming any governance role. Coinbase stood up an independent Quantum Advisory Council whose report is now the canonical source for the exposure figure: roughly 6.9 million BTC sit in outputs with exposed public keys, about 1.7 million of them in Satoshi-era pay-to-public-key outputs that can never be defended without moving the coins. A functioning post-quantum testnet exists. What does not exist is a single activated consensus change — and the freeze-the- vulnerable-coins debate (BIP-361) is where the real fight will happen. The binding constraint on Bitcoin’s quantum defense has migrated from physics to governance. That was Citi’s argument in May for why Bitcoin is more exposed than Ethereum, and six weeks of evidence has strengthened it.

Our posture: we nudge the code-breaking odds modestly upward — from roughly one-in-three within ten years toward something closer to 35–40% — and we are explicit about why we are not jumping further: 70 logical qubits is not 500, every advantage claim in history has faced classical catch-up, and the vendor roadmaps that now matter (2029–2030 fault tolerance) are marketing documents with a documented history of slippage. The three-clocks framework survives intact. The cryptography clock ticks slightly faster.

Two obligations before we close. A Bitcoin-benchmarked fund writing about a Bitcoin risk owes the reader the other side at full strength, so §7 argues Bitcoin's defense with nothing held back — it bounds the blast radius considerably, and stops exactly where the governance question begins. And a re-score is only worth publishing if it re-arms, so §9 ends the note the way v1.0 did: four new tripwires — three on the physics, one on Bitcoin's own progress — written down now, before the evidence arrives.

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1. The tripwire re-score

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We set three tripwires in June. One is warm.

From our v1.0 note (Jun 28, 2026): the pre-registered observations that would change our view of the quantum timeline — re-scored Aug 6, 2026

01	A sustained jump in logical-qubit counts, from dozens toward hundreds	● **PARTIALLY TRIGGERED** Frontier moved 48 → 70 (claimed) on Jul 30. One paper, one week old, classical-review cycle not yet run. Warm — not conclusive.
02	A fault-tolerant machine shipping ahead of the 2029 vendor targets	○ **NOT TRIGGERED** No machine early. But the 2029 roadmaps are holding, not slipping — and slippage was our base case.
03	A credible Shor demonstration against a non-trivial key	○ **NOT TRIGGERED** State of the art remains a 15-bit key (Apr 2026). Bitcoin's keys: 256 bits. The distance is exponential, not linear.

Scoreboard note: v1.0's marquee stat — zero economically useful quantum results, ever — now reads 0*. The asterisk is IBM's Jul 30 claim, and it resolves when the classical-simulation community takes its turn. We will score it publicly either way.

Pre-registration only means something if the scoring is mechanical — no reinterpreting the language after the fact. So we begin with the language exactly as we published it. v1.0 §7 said, verbatim: "What would change our mind: a sustained jump in logical-qubit counts (from dozens toward hundreds with low error), a fault-tolerant machine shipping *ahead* of the 2029 vendor targets, or a credible public demonstration of Shor's algorithm breaking a non-trivial key." Score each.

Tripwire 1 — logical-qubit counts: PARTIALLY TRIGGERED

When v1.0 published, the frontier was Quantinuum's Helios running 48 fully error-corrected logical qubits (November 2025). On July 30, 2026, IBM coordinated three announcements — with Algorithmiq, with Qedma, and with the University of Chicago — presenting all three as evidence that the field has entered “the quantum advantage era.” The three claims do not carry equal weight, and the distinction matters:

- The **IBM / University of Chicago** paper is the strong one: an explicit quantum-advantage claim backed by complexity-theoretic hardness arguments, a computation completed in about 15 minutes that the authors argue would take prohibitive runtimes classically — executed on **70 logical qubits** with a new error-correction method. (Verified: arXiv:2607.25941 — 70 logical qubits, 2,415 logical two-qubit operations, 468 logical T-gates.)
- The **Algorithmiq** result (simulation of a heterogeneous quantum material) and the **Qedma** result make more empirical claims — “the classical methods we tested became unreliable in this regime” — which is a weaker standard, and the fact-checking literature has already split the three claims along exactly this line.

Why “partially” and not “triggered”: the claim is one week old at this writing (published July 30); the pattern since 2019 is that classical simulation teams take weeks-to-months to respond, and they have caught every previous claim at least in part; and 70 is still “dozens,” not “hundreds.” The tripwire language said *sustained* jump. One paper is not sustained. But 48 → 70 at the frontier, inside a quarter, with an advantage claim attached, is precisely the shape of movement we said to watch. Status: warm.

For calibration: Google’s March 2026 resource estimate puts the break of ECC-256 — Bitcoin’s curve — at **fewer than ~1,200 logical qubits**. The frontier just moved from 48 to a claimed 70. The distance is a factor of ~17, down from ~25 at v1.0. That is what “partially triggered” means in numbers. (Verified against the primary: Google Research, Mar 31, 2026 — under 1,200 logical qubits / 90M Toffoli gates on under 500,000 physical qubits, minutes-scale runtime; arXiv:2603.28846.)

Tripwire 2 — fault tolerance ahead of 2029: NOT TRIGGERED (but the targets are holding)

No vendor has shipped a fault-tolerant machine early. What has changed is subtler and worth recording: the 2029-vintage roadmaps are, so far, *holding rather than slipping* — and roadmap slippage was our stated base case. IBM’s intermediate module (Kookaburra, the first processor module designed to store quantum information across modules) remains scheduled for 2026 delivery on the path to Starling in 2029. PsiQuantum, the most aggressive roadmap in the field, signed a $100 million CHIPS Act letter of intent in May — government money arriving on schedule is not proof of technical schedule, but it is the opposite of a funding stall. The July–August run of results (below) all de-risks pieces of the fault-tolerance stack rather than any single machine.

The score: not triggered, no machine early — but the *distribution* of roadmap outcomes shifted slightly toward “on time” and away from “slips two years,” and our v1.0 prior leaned on the slippage history. Status: white, with a note.

Tripwire 3 — a credible Shor demonstration: NOT TRIGGERED

The Q-Day Prize state of the art remains the 15-bit key broken in April 2026. Bitcoin's keys are 256 bits, and the difficulty is exponential in key size, not linear; 15 bits is a universe away, exactly as it was in June. No credible public demonstration of Shor's algorithm against a non-trivial key exists. Status: white. (Checked Aug 6 via secondary coverage: no larger break found. Re-confirm against Project Eleven's own leaderboard at publication — the site blocks automated fetch.)

And the marquee stat: "0 economically useful results" → "0, with an asterisk"

v1.0's opening number was zero — economically useful results a quantum computer has produced that a classical computer cannot. The IBM/Chicago claim is the first with the right *shape* to threaten that zero: a complexity-theoretic argument, logical qubits, verification framework, minutes-scale runtime. Our re-score is "0*": the asterisk stands for "one serious claim under review." What removes the asterisk in either direction is classical catch-up — either the simulation community matches the result within months (asterisk deleted, zero restored, as with every prior claim), or it doesn't (the zero flips, and that is a genuinely new world). We commit now to scoring this publicly in the next update either way.

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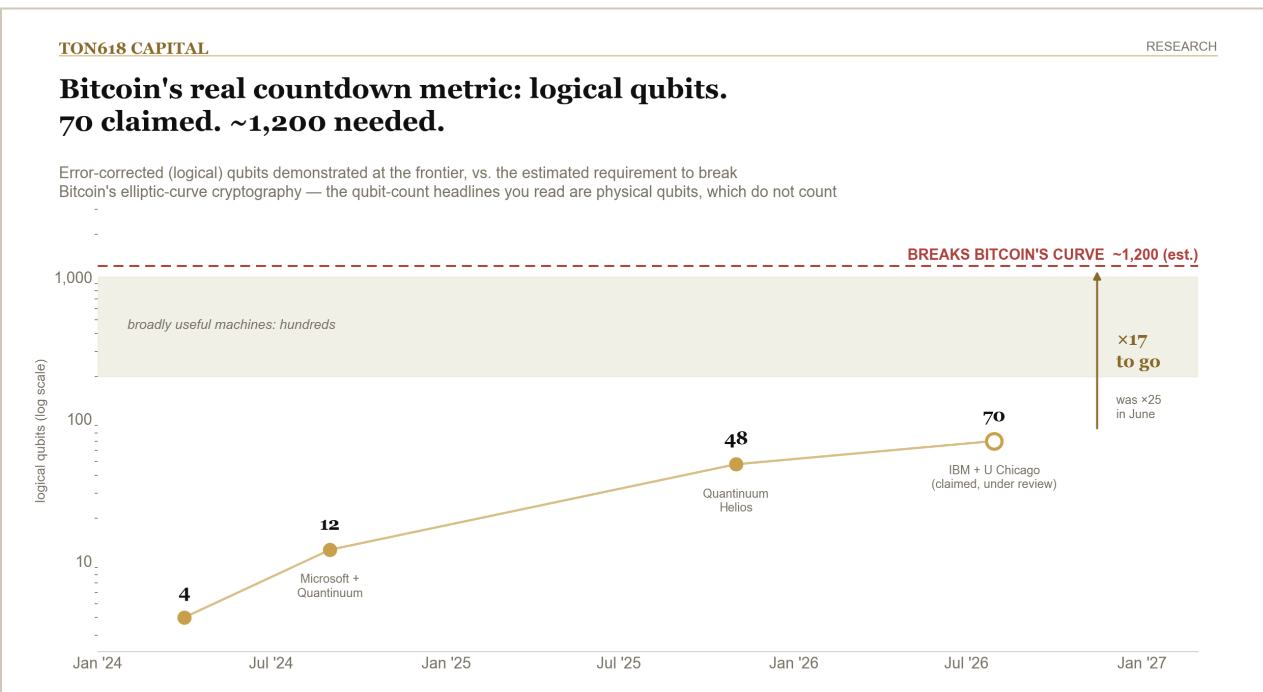
2. Six weeks of science: the densest run since Willow



Frontier demonstrations of fully error-corrected logical qubits; IBM figure as claimed Jul 30, 2026, pending classical-simulation review. Requirement estimate per 2026 research; sourcing in the note appendix.

Between v1.0's publication and this note — 39 days — the field produced four Nature-tier results and one coordinated advantage announcement. For pacing purposes, the cadence is the datum; no single result decides anything, but the clustering is unlike any six-week window since December 2024:

- **July 15 — Quantinuum + academic partners:** first universal *topological* gate set, using non-Abelian anyons on the H2 processor — a 54-qubit entangled state realizing S3 non-Abelian symmetry, published in Nature (Jul 15; DOI 10.1038/s41586-026-10709-y — press coverage ran Jul 17). Topological encodings are the "protected by geometry" route to error resistance; a universal gate set is the checkbox that matters.
- **July 29 — HRL Laboratories:** an 18-qubit silicon spin QPU operating *autonomously at 4 kelvin* with a custom cryogenic CMOS control chip, in Nature. Small qubit count; large implication — control electronics living at cryogenic temperature next to the qubits is one of the two or three engineering walls between hundreds of qubits and millions (Nature, Jul 30, 2026 issue; HRL release Jul 29). Concentration disclosure a careful reader deserves: IBM signed a definitive agreement to acquire HRL on July 23 — six days before this paper — which means IBM has a hand in four of the five headline results of the window. The cadence is real; its independence is thinner than the mastheads suggest.
- **July 30 — IBM × 3:** the advantage-era announcements discussed in §1.
- **August 5 — D-Wave:** a fast (~500 ns), high-fidelity (~99.9%) two-qubit entangling gate on a superconducting *dual-rail* qubit architecture with native hardware-level error detection, published in Nature — D-Wave's first serious gate-model credential, with simulations suggesting the architecture could cut logical error rates by ~10× per error-correction increment (D-Wave release / Nature, Aug 5; the 10× figure is D-Wave's own simulation, and is labeled as such).

Read against v1.0's framework: none of this produces commercial value today, and all of it is exactly the kind of *below-the-threshold plumbing* — gates, control, encodings, verification — that a field crosses on the way from "threshold crossed" to "machine exists." In v1.0 we wrote that the foundation was finally real. The update is that people are now building on the foundation, quickly, in public, in the peer-reviewed literature rather than the press-release channel. That is what acceleration looks like from the outside.

And while all of this was printing, the stocks that supposedly price this future were doing precisely the opposite.

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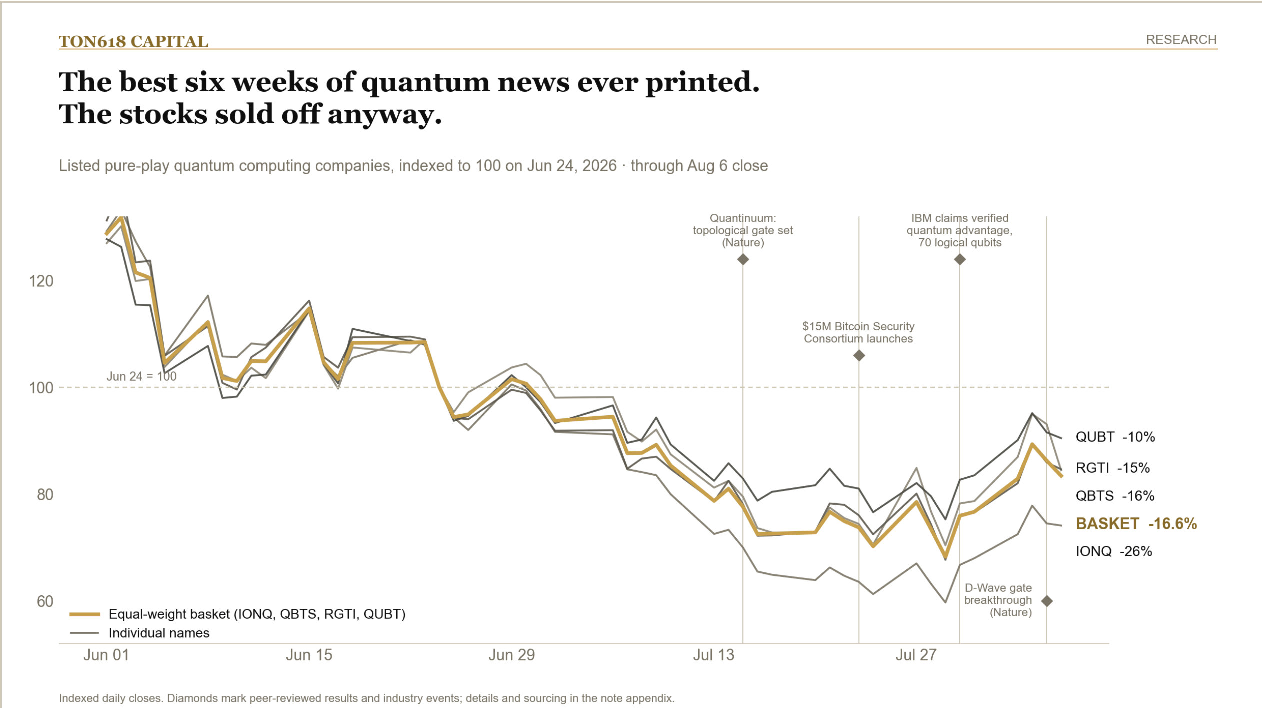
3. The stocks: not a breakthrough signal — a hangover



If a breakthrough were truly close, the market should smell it first — that is the efficient-markets intuition, and it deserves a test rather than an assumption. Here is the test: **the quantum stocks are not signaling an imminent breakthrough; they are signaling the unwind of a speculative episode, and they de-rated straight through every result in §2.**

Priced off the v1.0 note's own pricing date (June 24, 2026) through the August 6 close:

Ticker	Company	Jun 24 close	Aug 6 close	Change	Mkt cap (Aug 6)
IONQ	IonQ	$53.60	$39.72	**−25.9%**	$14.8B
QBTS	D-Wave Quantum	$22.98	$19.41	**−15.6%**	$7.2B
RGTI	Rigetti Computing	$19.53	$16.53	**−15.4%**	$5.5B
QUBT	Quantum Computing Inc.	$9.70	$8.78	**−9.5%**	$2.0B
—	Equal-weight basket	100	≈83.4	**−16.6%**	~$29.5B combined
ARQQ	Arqit (post-quantum security)	$28.64	$21.47	**−25.0%**	—

From the early-June highs the drawdowns are deeper still: IonQ −43% from its June 1 close, Rigetti −36%, D-Wave −33%, QUBT −29%. And the single most instructive tape print of the window: **D-Wave published a genuine Nature hardware breakthrough on August 5 and fell roughly 9% the following session** — a sell-the-news reaction in which coverage framed IBM's and Google's gate-model progress as a competitive *threat* to D-Wave rather than a rising tide. When a Nature paper is a down-9% event, the marginal buyer is not doing physics.

Three observations for the permanent record:

1. **The de-rating and the news flow are causally disconnected.** The slide began in early June, weeks before the July results, and did not pause for any of them. This is the December–January speculative episode unwinding on its own financial-market clock — rate expectations, risk appetite, supply of new shares — not a verdict on the science.
2. **Even the defensive side fell.** v1.0 suggested the durable trade might be the post-quantum-security vendors, whose revenue is driven by government migration deadlines (2030–2035) far more certain than Q-Day. Arqit fell 25% anyway. Deadline-driven revenue is a fundamentals argument, and this tape was not trading fundamentals in either direction.
3. **Therefore: do not read equity prices as a Q-Day sensor — in either direction.** The same people citing the 2025 melt-up as evidence the breakthrough was near would now have to cite this drawdown as evidence it receded. Neither inference was ever valid. The sensor that works is the one in §1: logical-qubit counts in the peer-reviewed literature.

Valuation postscript: the multiple is compressing from both directions — prices down ~17% while revenue accelerates (IonQ's Q2 print: $80.0M, +287% YoY, nearly half of v1.0's combined four-company TTM figure in a single company-quarter). The basket remains priced off a 2035 outcome — cheaper than June, not cheap — but the honest multiple is now several fold lower than v1.0's 268×, and the final note must re-run it on fresh 10-Qs rather than reprint the stale figure. The v1.0 barbell framing (platform incumbents and suppliers as the core, pure-plays as deliberately-sized lottery tickets, the PQC/consultancy migration trade as the sleeper) survives unchanged. [OPEN: refresh combined TTM revenue from latest 10-Qs at publication — IonQ Q2 confirmed at $80.05M (+287% YoY), others pending. Share-count basis: verifier independently reproduced both the ~$52B (v1.0 point-in- time) and ~$37B (current share count) figures for Jun 24; which basis to print is an authorial call, still open.]

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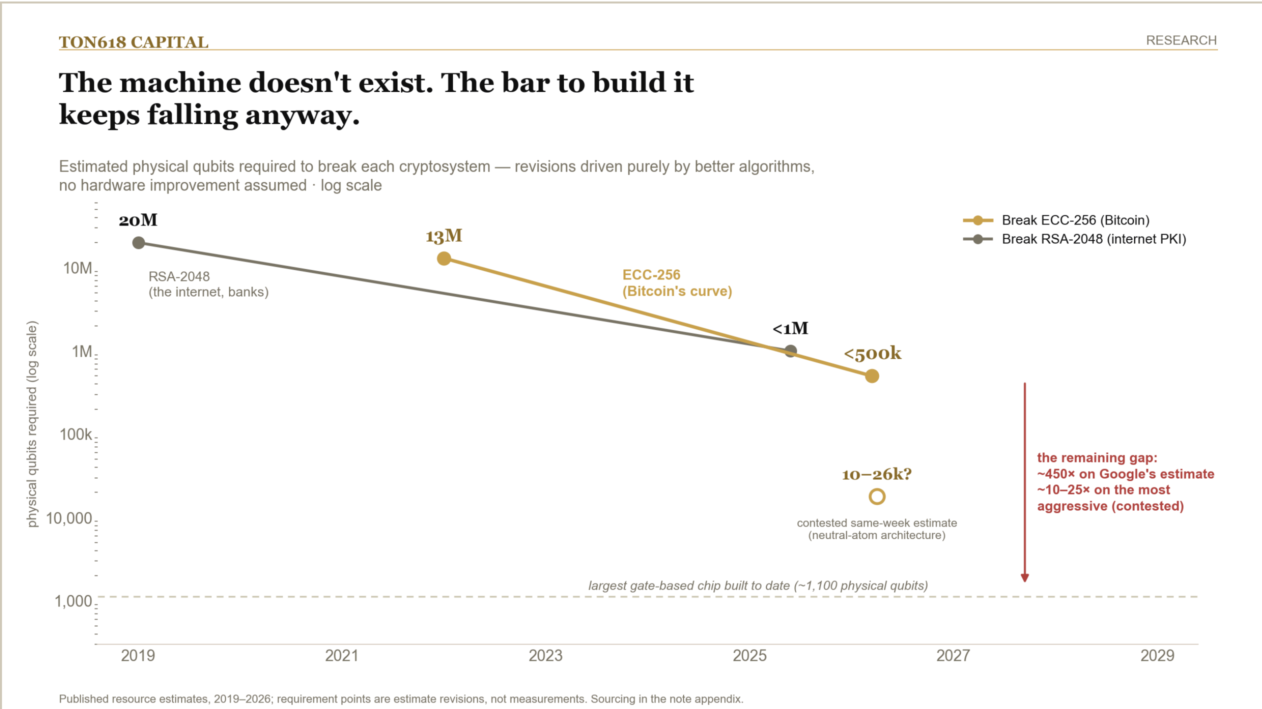
4. Is the timeline accelerating? Three lines of evidence, one answer



Strip the tape out, then. What remains is the evidence that binds: the published resource estimates, the demonstrated hardware, and the behavior of the people whose job is to be right about this. Three lines, then the synthesis.

Line 1 — the resource estimates keep falling. This was v1.0's most striking chart and it has a new point. In 2019 the canonical estimate for breaking RSA-2048 was ~20 million physical qubits; Gidney's May 2025 revision cut it to under 1 million — 20× from algorithms alone. The new datum is Bitcoin-specific: Google's March 2026 work puts an attack on ECC-256 at **fewer than ~1,200 logical qubits and under 500,000 physical qubits, with runtimes measured in minutes** on a future fault-tolerant machine. For reference, the 2022 academic estimate (Webber et al.) for breaking a Bitcoin key inside a day was ~13 million physical qubits. The target is walking toward the hardware at roughly an order of magnitude every three to four years, with no hardware improvement required. (All figures verified against primaries: Gidney–Ekerå 2019; Gidney 2025, arXiv:2505.15917; Webber 2022, AVS Quantum Science; Google 2026, arXiv:2603.28846.)

An honest complication, flagged before a reader flags it for us: Google's number is not even the aggressive end of the same-vintage literature. A Caltech/Oratomic estimate published the same week (March 2026, neutral-atom architecture, different and contested assumptions) puts an ECC-256 break at 10,000–26,000 physical qubits — twenty to fifty times below Google's figure. We treat Google's as the working estimate because its assumptions sit closer to the architectures actually being scaled today, but the spread is the point: the live range of expert estimates now runs from ~10,000 to ~500,000 physical qubits, and even its conservative end keeps falling. The direction of travel is not in dispute; only the speed is.

Line 2 — the builders moved their language, then moved their claims. IBM's CEO predicted the first real-world quantum advantage would land in 2026 (April, on the record); IBM then claimed it (July 30). The same CEO, in the same July 30 CNBC appearance, put crypto-relevant capability at three to four years — the most aggressive timeline any major-vendor executive has attached to the cryptography milestone. Nobel laureate John Martinis — who built Google's early superconducting program — spent the spring arguing Bitcoin will be among the *first* real-world targets, because a derived private key monetizes instantly and irreversibly. Discount executive timelines as marketing, as we always do; but note that in June the marketing said "advantage soon," and by August the papers said "advantage claimed." The gap between talk and print shrank, and that gap is itself a pacing indicator.

Line 3 — the deadline-setters are not waiting. Google's internal post-quantum migration targets 2029. NIST's horizon remains 2035. Coinbase's CEO works to 2029. The U.S. government PQC deadlines from v1.0 stand. Nobody with fiduciary exposure to being wrong has relaxed a deadline in six months; several have tightened operational work against them (§5).

The synthesis. v1.0 put a code-breaking machine at roughly one-in-three within ten years, up from one-in-five a year earlier. On the evidence above we mark that to **~35–40% within ten years** — a nudge, not a jump, and we want to be explicit about the reasons for restraint, because they are load-bearing: (a) 70 logical qubits is a claim one week old from one group, not a sustained frontier; (b) every previous advantage claim has been at least partially matched classically, and the response cycle has not run; (c) the remaining engineering wall — from tens of logical qubits to the ~1,200 that threaten ECC-256, with the required gate depth and magic- state throughput — is precisely the part of the problem that has never been demonstrated at any scale; and (d) vendor roadmaps holding for six months is weak evidence against a slippage base rate built over fifteen years. The three-clocks framework from v1.0 is unchanged: narrow scientific value is arriving on schedule (arguably early), commercial value remains a 2030s story, and the cryptography clock — the only one Bitcoin cares about — now ticks slightly faster than it did in June.

What would make the next update a *jump* rather than a nudge is pre-registered, as before — the reset tripwires are formalized in §9, and they close the note.

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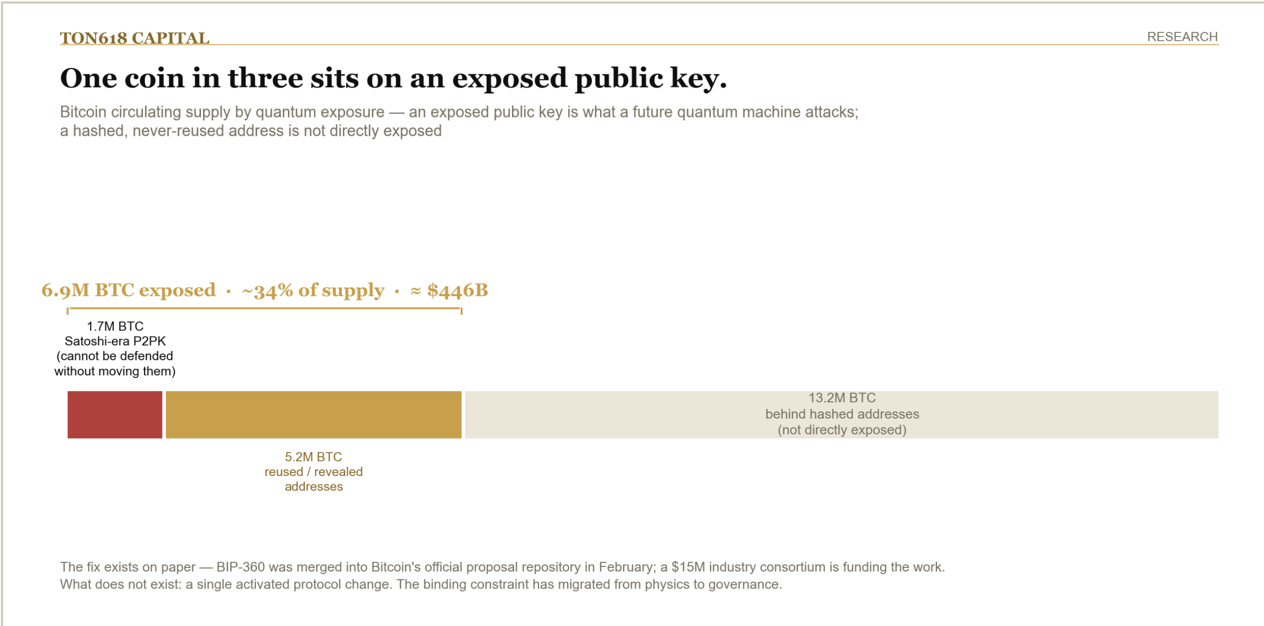
5. What Bitcoin is actually doing about it: funded, specified, not activated



v1.0 treated Bitcoin's quantum exposure as a monitorable tail risk and listed the BIP-360/361 process as the live signal. Six weeks — and one busy summer — later, here is the tangible ledger. It is more substantial than we expected, and it has a hole in the middle.

The specification is real. BIP-360 — Bitcoin's first formal post-quantum proposal — was merged into the official BIPs repository on February 11, 2026. Along the way it was redesigned from the original Pay-to-Quantum-Resistant-Hash (P2QRH, June 2024) into **Pay-to-Merkle-Root (P2MR)**: the output commits directly to a script-tree Merkle root with no internal key at all, removing the key-path exposure entirely and giving future post-quantum signature schemes a clean structural base to plug into. Merge into the repo signals documentation standards met and formal discussion open — not endorsement, not activation. But a real, reviewed, evolving specification now exists where two years ago there was a mailing- list thread.

The money is real, and correctly humble. On July 23, nine institutions — BlackRock, Coinbase, Strategy, Fidelity Digital Assets, Galaxy, Block, Blockstream, Anchorage Digital, and ARK Invest — launched the **Bitcoin Security Consortium**: $15 million pledged over three years to fund open-source developers and researchers working on the network's long-term security, with members directing funds independently and the consortium taking *no role in governance or protocol decisions*. Read both halves. The signatory list is the institutional Bitcoin complex, essentially in full — the ETF issuer, the largest corporate holder, the largest exchange, the custody banks. And $15 million against a ~$446 billion exposed-coin problem is a signal, not a solution: roughly what one of these firms spends on a Super Bowl ad, aimed at a problem they all describe as existential-if- mishandled. The correct reading is that the consortium exists to fund engineers without touching governance — because governance is the part nobody can buy.

The engineering is real. Coinbase has dedicated engineering headcount to BIP-360 and the broader migration path, and stood up an independent **Quantum Advisory Council** whose report is now the canonical exposure census: approximately **6.9 million BTC in UTXOs with exposed public keys**, including about **1.7 million BTC in Satoshi-era pay-to-public-key outputs** — coins that cannot be defended by any protocol change short of freezing or moving them, because their public keys have been on-chain for fifteen years. BTQ Technologies' Bitcoin Quantum testnet (v0.3.0, March 2026) provides the first functional validation of the P2MR design running end-to-end. A post-quantum migration proposal with a phased structure (new-output support → deprecation of vulnerable spends → the freeze question) is under active discussion on bitcoindev.

The hole in the middle: activation. Zero consensus changes have activated. None is scheduled. And the hard question — BIP-361's territory, what to do about coins that will never migrate, including Satoshi's — is not an engineering question at all. Freeze them and you have retroactively confiscated property to defend a principle; leave them and you have accepted that roughly a third of the supply becomes a bug bounty the day a machine exists, with the fire-sale and confidence dynamics that implies. Ethereum, with a living founder, a research culture organized around hard forks, and a coordinated migration already underway, does not face this shape of problem. This was Citi's May argument for why Bitcoin's quantum exposure exceeds Ethereum's despite identical curve cryptography — **the binding constraint is governance, not physics** — and everything that happened this summer is consistent with it. The money and the code arrived. The decision has not.

Updated monitoring list for the fund (supersedes v1.0 §7's list): 1. Logical-qubit frontier in the peer-reviewed literature (currently: 48 sustained / 70 claimed; alarm at ~200 sustained). 2. Classical response to the IBM/Chicago claim (score it publicly in the next update). 3. BIP-360 reference implementation progress and any activation-signaling from Core maintainers (currently: none). 4. The BIP-361 freeze debate — the first credible activation timeline for *any* PQ output type is the event; the freeze fight is the risk. 5. Share of supply migrated to hash-protected outputs (currently ~6.9M BTC exposed; falling exposure = the system working). 6. Q-Day Prize key size (currently 15 bits). 7. Consortium funding actually deployed vs. pledged (currently $0 visibly deployed; first grants are the tell).

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6. The Street picks a side (new since v1.0)

While the engineers built, the Street argued. v1.0 noted only that BlackRock had added a quantum risk factor to the IBIT filing; the intervening months produced a full-blown allocator debate, and it sorts almost perfectly by book:

Acted: Christopher Wood (Jefferies, global head of equity strategy) — removed his entire 10% bitcoin model-portfolio allocation in January, splitting it into physical gold and gold miners, calling quantum an existential threat to the store-of-value thesis and citing the 4–10M BTC exposure range. The single most significant datum in this section: an early institutional bull (allocated since 2020) who reversed on this specific risk.

Warned: Citi (Alex Saunders, May) — bitcoin more exposed than Ethereum on governance grounds; ~6.7–7M BTC exposed. Chamath Palihapitiya — bitcoin as the "honeypot": the first target of non-state actors because a broken key monetizes instantly. Nobel laureate John Martinis — early target, on the record repeatedly this spring.

Bounded: Bernstein (Gautam Chhugani, April) — "neither existential nor novel"; a manageable 3–5-year upgrade cycle; mining safe; 1.7M Satoshi-era coins the real exposure. Benchmark (Mark Palmer, January) — "real but distant." ARK Invest (March) — long-term risk, not imminent. BlackRock — risk factor filed, but research framing quantum as one of the last "walls of worry," with successful migration a potential *strengthening* event.

Dismissed: Mike Novogratz (Galaxy) — the network adapts; the real risk is developer discord over the fix (note: this is the governance argument wearing an optimist's coat). Michael Saylor (Strategy) — a decade-plus away; warns against premature protocol changes (note: Strategy signed the consortium anyway — watch what they fund, not what they say).

The pattern to flag for readers: with the single exception of Wood, the urgency of every participant's assessment predicts their positioning almost perfectly. The longs find it manageable; the unexposed find it existential; the gold-adjacent find it terminal. Wood is the exception that proves the diagnostic value of the exercise — he changed his book to match his assessment rather than his assessment to match his book. That is what makes his January move the one worth studying, whatever one thinks of its conclusion.

(House note, internal consistency: per fund doctrine this note states no BTC expected return and no price implication of any scenario. The quantum question enters the fund's process as a monitored tail risk and a tripwire list, §5, nothing more.)

7. The steelman: Bitcoin's defense, at full strength

Section 6 ends on an uncomfortable observation: nearly everyone's quantum verdict predicts their book. We are a Bitcoin-benchmarked fund writing about a Bitcoin risk, which means the same critique aims at us — from both directions. The only defense is to argue the strongest version of the case we did *not* lead with, and let the tripwires adjudicate. Here is Bitcoin's defense with nothing held back.

1. Most of the exposure can defend itself today, with no fork at all. The 6.9M BTC figure is a stock, not a fate. Roughly 5.2M of it sits in reused or otherwise revealed addresses whose owners can move to a fresh, hashed address this afternoon — an ordinary transaction, no protocol change, no governance fight, no consortium. The exposure number is therefore partially self-liquidating as the threat becomes salient: rational holders migrate on their own clock, and a *falling* exposed-supply series (§5's monitor #5) is the system working without anyone's permission. What cannot be defended this way is the ~1.7M BTC in Satoshi-era P2PK — the coins with no living owner. The steelman shrinks the problem from "a third of the supply" to "the dead coins"; it does not dissolve it.

2. Hashed addresses stay safe until the moment they spend. A never-reused modern address exposes its public key only in the seconds-to- minutes between broadcast and confirmation. Attacking that window requires a machine that runs Shor in minutes — a far taller order than the years- long at-rest attack on already-exposed keys. The practical consequence is underappreciated: Bitcoin's migration can succeed even if it starts *late*, because the majority of supply is not racing the first CRQC; it is racing the much stronger machine that comes years after.

3. Governance latency is not governance inability — and slowness cuts both ways. The network that supposedly cannot upgrade shipped P2SH, SegWit (2017), and Taproot (2021): a consensus change roughly every four years, which is precisely the cadence Bernstein's 3–5-year window requires. Ossification is not a bug the quantum threat exposes; it is the security model that makes a $2 trillion bearer asset possible at all — and it protects the migration itself. A chain that could be rushed into a cryptography swap by a news cycle could also be rushed into a bad one. The governance-constraint critique (Citi's, and ours in §5) is real, but it describes the *freeze decision*, not the upgrade: adding a post-quantum output type is additive and uncontroversial; only the fate of the dead coins is a fight.

4. On Q-Day, everything breaks — and Bitcoin is among the few things that can be rebuilt in place. A CRQC does not selectively attack secp256k1. It breaks TLS, code-signing, interbank messaging, custody infrastructure, and every archived signature on earth. Bitcoin can hard-fork its cryptography; a land registry, a signed treaty, or a decade of recorded bank traffic cannot retroactively re-sign itself. BlackRock's framing is the institutional version of this point: quantum is one of the last "walls of worry," and a *completed* migration would remove it — the asset that crossed the post-quantum bridge first, in public, gains a property nothing else in the legacy system can demonstrate.

5. The threat is unusually observable — Bitcoin must beat the machine, not the headline. The countdown metric (logical qubits in the peer-reviewed literature) is public, vendor roadmaps are marketing documents published years ahead, and the national-security world's own migration deadlines (2029–2035) bracket the consensus timeline. Very few tail risks announce themselves this loudly. The window in which acting is possible is wide open and well-lit.

6. Attacker economics are, at minimum, contested. Stealing a million coins collapses the value of the loot as it moves; a rational state actor with the first CRQC has quieter and richer targets than crashing a transparent ledger. We flag the honest counter in the same breath: the honeypot argument (Palihapitiya's) says irreversibility and instant monetization make Bitcoin uniquely attractive to *non-state* actors who don't care about preserving the system — and a public quantum theft anywhere collapses confidence everywhere, whatever the attacker's P&L. This argument reduces the probability of the worst path; it does not retire it.

Where the steelman stops. Taken together, the defense bounds the blast radius — from "a third of supply" to the dead coins, and from "race the first machine" to "race the stronger machine" — and it establishes that the *upgrade* is well inside Bitcoin's demonstrated governance capacity. What it cannot do is answer the freeze question, because that question is not technical, and no amount of engineering funding buys the answer. Our assessment stands, both halves: the risk is manageable on the evidence, and the remaining constraint is governance. The tripwires below are how we avoid arguing either side of that on vibes.

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8. Investment relevance & the TON618 read (updated)

For the fund, six weeks of evidence changed the monitoring, not the position:

- **The barbell survives the drawdown.** Platform incumbents and their suppliers remain the core way to own the theme; the pure-plays remain lottery tickets that just got ~17% cheaper and no more certain; the post-quantum migration vendors remain the fundamentals trade, now also cheaper, with their deadline-driven revenue thesis untouched by a tape that was not trading fundamentals.
- **The asymmetry still cuts the right way.** A credible acceleration damages every RSA/ECC-dependent incumbent — the entire banking, government, and internet stack — at least as much as it damages a Bitcoin network that now has a specification, a testnet, and a funded engineering pipeline. What Bitcoin uniquely owns is the governance risk (§5), and that is the correct thing for a Bitcoin-benchmarked fund to monitor.
- **The monitoring list is §5's.** The next scheduled re-score publishes when either the classical-response verdict on IBM/Chicago is in, or a tripwire triggers, whichever is first.

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9. The new tripwires — set August 2026

TON618 CAPITAL RESEARCH

The tripwires, re-armed.

Set Aug 6, 2026 — the pre-registered observations that will move our view before the next update.
Three on the physics, one on Bitcoin itself. Written down before the evidence arrives.

T1 The refutation clock — *resolves by Jul 2027*

IBM's 70-logical-qubit advantage claim either survives twelve months of classical attack — or it doesn't. Survives → the scoreboard's zero finally flips. Matched → the zero is restored.

T2 A sustained frontier of ~200+ logical qubits — *open*

Two independent groups, error rates consistent with deep circuits. The geometric midpoint between today's ~70 (claimed) and the ~1,200 estimated to break Bitcoin's curve.

T3 A Shor demonstration that scales — *open*

A public break of a 64-bit-or-larger elliptic-curve key. Today's record: 15 bits — a toy a laptop can simulate. At 64 bits, the road to 256 becomes engineering, not undemonstrated physics.

T4 Bitcoin's activation signal — two-sided — *open*

Positive: a post-quantum output type gets a real activation path, or exposed supply (~6.9M BTC) starts falling. Negative: an activation attempt visibly fails — proof the governance constraint binds.

The standing commitment, unchanged since v1.0: tripwires are written before the evidence, so the next update is a score — not a story.

v1.0 closed with three tripwires and this note exists because we went back and scored them. Same discipline, re-armed. Four this time — three on the physics, and one on Bitcoin itself, because §5's conclusion (the binding constraint is governance) obliges us to watch a governance metric and not only a laboratory one. Each is written to be scoreable by a stranger.

T1 — The refutation clock. The IBM/University of Chicago advantage claim (Jul 30, 2026; 70 logical qubits) either survives twelve months of classical-simulation attack or it does not. Survives materially intact by July 2027 → our marquee stat flips from 0* to 1, and the next update is a jump, not a nudge. Substantially matched classically → the asterisk is deleted, the zero is restored, and we will print that with equal prominence. This one resolves on a known clock.

T2 — A sustained logical-qubit frontier of ~200+. Not one paper: at least two independent groups demonstrating on the order of 200 or more logical qubits with error rates consistent with deep circuits. That is the “dozens toward hundreds” language of v1.0 with the numbers filled in, set roughly at the geometric midpoint between today’s frontier (~70 claimed) and the ~1,200 estimated to threaten ECC-256.

T3 — A Shor demonstration that scales. A public break of an elliptic- curve key of 64 bits or more. Today’s 15-bit prize-winners are toy instances a laptop can simulate; a 64-bit break is the approximate threshold at which the remaining distance to 256 bits becomes an engineering roadmap — more qubits, more time — rather than undemonstrated physics. The specific number is our choice and we state it so we cannot move it later.

T4 — Bitcoin's activation signal (new, and two-sided). Triggers positive: any post-quantum output type reaches a concrete activation mechanism — a proposed activation path with maintainer support and ecosystem signaling — or the measured exposed-supply series (start: ~6.9M BTC) begins falling at a rate consistent with deliberate migration. Triggers negative: an activation attempt is made and visibly *fails* — the first real evidence that the governance constraint binds in practice, not just in argument. Either direction re-scores §5; silence scores as silence.

The standing commitment is unchanged from v1.0: write it down before the evidence arrives. The lab will keep doing what it does; the tape will keep doing what it does; the chain will decide, or it won't. When any of them crosses a line above, the next update will say so the way this one did — as a score, not a story.

Sources & Method (appendix — becomes the note's appendix; charts carry no source captions per house rule)

Market data (verified directly, Massive Market Data API, closes as of 2026-08-06): IONQ, RGTI, QBTS, QUBT, ARQQ daily aggregates Jun 1–Aug 6 2026; market caps = Aug 6 close × current share counts (reference data). Basket = equal-weight average of the four pure-play indexed closes, base 100 = June 24, 2026 (v1.0 pricing date).

Science and industry (press-sourced; primaries verified in the Aug 6 adversarial pass — see Verification status below): - IBM advantage claims: IBM newsroom (Jul 30, 2026); Decrypt; postquantum.com fact-check of the three papers; Dataconomy (Aug 3). - IBM Krishna 2026-advantage prediction: The Quantum Insider (Apr 30, 2026). - Quantinuum topological gate set: Quantinuum blog / Nature (Jul 17, 2026). - HRL silicon QPU: Nature (Jul 29, 2026) via Quantum Computing Report. - D-Wave dual-rail gate: D-Wave press release / Nature (Aug 4-5, 2026). - PsiQuantum CHIPS award: Quantum Zeitgeist (May 2026). - Google ECC-256 resource estimate: March 2026, via postquantum.com migration roadmap; Google PQC target 2029; NIST horizon 2035. - Gidney RSA-2048 estimates (2019: 20M; 2025: <1M): v1.0 sourcing. - Webber et al. 2022 ECC estimate (~13M physical/24h): AVS Quantum Science.

Bitcoin workstream: - BIP-360 merge (Feb 11, 2026) and P2QRH→P2MR history: Datawallet explainer; bitcoindev list; TFTC. - Bitcoin Security Consortium (Jul 23, 2026, nine members, $15M/3yr): CoinDesk; Strategy press release; TFTC. - Coinbase Quantum Advisory Council; 6.9M BTC / 1.7M P2PK exposure: Coinbase blog. - BTQ Bitcoin Quantum testnet v0.3.0 (Mar 2026): postquantum.com. - BIP-361 freeze debate: KuCoin explainer; bitcoindev list.

Allocator debate: Bloomberg / The Block / CoinDesk (Wood, Jan 16); CoinDesk (Citi, May 18; Bernstein, Apr 8; Benchmark, Jan 29; ARK, Mar 12; Novogratz, Feb 3; Martinis, Apr 7); Forbes (Armstrong, Apr 3); CCN (Carter); cryptonews (Palihapitiya). Sell-side notes are paywalled; figures are as reported in coverage and marked approximate.

Verification status (adversarial claim-check run Aug 6, 2026): Independent verifier reproduced every market-data figure (closes, changes, caps, basket, drawdowns — including QBTS's −9.28% on Aug 6) via a second data source, confirmed all arXiv/Nature science claims against primaries (IBM/Chicago arXiv:2607.25941; Google arXiv:2603.28846; Quantinuum Nature DOI 10.1038/s41586-026-10709-y; Gidney arXiv:2505.15917; Webber AVS Quantum Science 2022), and confirmed the BIP-360/consortium/Coinbase workstream, all allocator attributions, and the Condor gate-based record. Corrections applied from that pass: exposed-coin value restated \$460B → ~\$446B (6.9M × BTC ≈ \$64.65k, Aug 6); Quantinuum Nature date Jul 15 (press Jul 17); Caltech/Oratomic same-vintage estimate added to §4; IBM–HRL acquisition (Jul 23 agreement) disclosed in §2; PsiQuantum award noted as a letter of intent; IonQ Q2 revenue (\$80.05M, +287% YoY) folded into §3.

Still open before publication: 1. Share-count basis for v1.0's \$52B vs. recomputed \$37B (both reproduced by the verifier; which to print is an authorial call). 2. Combined TTM revenue refresh from the remaining Q2 10-Qs. 3. Re-confirm Project Eleven's leaderboard directly (site blocks automated fetch; secondary coverage shows no break beyond 15 bits as of Aug 6). 4. The Aug 3–4 bounce (IONQ +7.4% Aug 4) and QBTS's −9.3% on Aug 6 — check for news vs. beta before characterizing.

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